



Roll No:

BTECH**(SEM. VI) THEORY EXAMINATION 2025-26
ELECTRIC & HYBRID VEHICLES TECHNOLOGY****TIME: 3 HRS****M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question
a.	What is a Hybrid Electric Vehicle?
b.	Write down any two advantages of electric vehicles.
c.	Define fuel efficiency of a vehicle.
d.	Define power flow control in hybrid vehicles.
e.	What is meant by motor drive control?
f.	Define communication system in electric vehicles.
g.	What is meant by energy storage technology?

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Q no.	Question
a.	Write down the major components of hybrid electric vehicles and explain their functions.
b.	Explain braking fundamentals and regenerative braking systems used in electric vehicles.
c.	Describe the working principle of Switched Reluctance Motor drives.
d.	Discuss different energy storage technologies used in electric vehicles.
e.	Explain the advantages and limitations of flywheel energy storage systems.

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

a.	Discuss the working principle, advantages, disadvantages, and applications of Fuel Cell Electric Vehicles (FCEVs).
b.	Explain battery technologies used in electric vehicles and compare their characteristics.

4. Attempt any one part of the following:**07 x 1 = 07**

a.	Explain in detail the mathematical models used to describe vehicle performance with relevant equations.
b.	Discuss transmission characteristics and their importance in electric and hybrid vehicles.

5. Attempt any one part of the following:**07 x 1 = 07**

a.	Explain in detail the electric components used in Hybrid and Electric Vehicles with neat block diagram.
b.	Discuss the construction and working of Permanent Magnet Motor drives.

6. Attempt any one part of the following:**07 x 1 = 07**

a.	Explain the integration of electric machine with ICE in hybrid vehicles.
b.	Describe the factors considered in sizing the propulsion motor.

7. Attempt any one part of the following:**07 x 1 = 07**

a.	Explain the features and working principle of ultra-capacitors.
b.	Discuss different types of fuel cells used in EVs.



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